SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO5 – Articulation	**Assessment:**	Assessment Tool 1 – ConstraintBased Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
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Section / Batch:	BE CSE AI / 2024	**Date:**	14-08-2026

Code of Conduct

*I **NARAPPAGARI SRAVYA(192472192)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:SRAVYA

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2

Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

Annexure A – Constraint-Based Problem Solving

1. **Given the following paragraph:**

   "John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

   Tasks: a) Identify all referring expressions (mentions) and possible antecedents.

   b) Apply the following **constraints** to resolve the references:

- Gender and number agreement
- Recency (prefer recent antecedents)
- Semantic compatibility (e.g., animacy and action suitability) Coherence (maintain consistent entity chains)

c) Draw a **constraint graph** or table showing variables, domains, and which constraints eliminate possible links.

d) Provide the final resolved coreference chains and rewrite the paragraph with clear references.

e) Justify the priority order of your constraints and discuss what would happen if the recency constraint is relaxed.

2. **Conversation History:** User: “I have an important exam tomorrow but I’m not able to concentrate.”

   Required Dialog Act: Advise + Encourage (Give advice and motivate the student).

   Constraints:

1. Maintain **entity coherence** (reuse “exam”, “concentrate”, “you”).
2. Use **discourse relations** (Cause-Effect or Elaboration).
3. Include at least **two** of these keywords: focus, break, confident.
4. Response length: 2–3 sentences only.
5. Logical consistency (no contradictory statements).
6. Politeness and positive tone.

   Tasks: a) Identify the relevant discourse planning steps needed before generation.

b) Generate **three** different possible responses that satisfy **all** the above constraints.

c) Evaluate the three responses and select the best one. Justify your choice using coherence and constraint satisfaction criteria.

d) Explain how violating any two constraints would affect the quality of the dialog.

3. **Source Sentence:** "The bank by the river flooded after the storm, but it was saved by quick action."

   Note: "bank" is ambiguous (financial institution or riverbank).

   Constraints:

1. Correct **Word Sense Disambiguation** using context.
2. Create a **predicate logic** representation (e.g., using predicates like flood(x), location(x,y)).
3. Maintain **discourse structure** (Contrast relation between the two clauses).
4. Preserve all important entities and relations (no information loss).
5. Ensure **text coherence** in the generated output.

   Tasks: a) Resolve the ambiguous word "bank" and justify using constraints.

b) Write a predicate logic representation of the sentence.

c) Generate a good **target language sentence** (in any Indian language or simple English paraphrase) that satisfies all constraints.

d) Draw a simple **discourse tree** (RST-style) showing the rhetorical relation.

e) Discuss the advantages of using constraint-based approach over pure statistical machine translation for this sentence

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritizatio n	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1							
2							
3							

Faculty In-charge	Course Coordinator	Program Director
Signature: ________________	Signature: ________________	Signature: ________________
Date: ____________________	Date: ____________________	Date: ____________________

1. Coreference Resolution

Given paragraph:

"John and Mary went to the park. He brought a ball. She wanted to play with it. The dog chased him excitedly. Finally, they all went home."

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	John, Mary
She	John, Mary
it	Ball, park
him	John, Mary, dog
they	John, Mary, dog

b) Applying the Constraints

1. Gender and Number Agreement

- **He → John** because John is male and singular.
- **She → Mary** because Mary is female and singular.
- **it → ball** because "ball" is singular and non-human.
- **him → John** because "him" is masculine and refers to a person.
- **they → John + Mary + dog** because "they" is plural.

2. Recency

Recent mentions are preferred, but recency should not override grammatical agreement.

3. Semantic Compatibility

- John can **bring a ball**.
- Mary can **want to play with a ball**.
- A dog can **chase a person**.
- John and Mary together with the dog can **go home**.

4. Coherence

The references should form a consistent chain throughout the paragraph.

c) Constraint Table

Variable	Initial Domain	Constraints	Final Choice
He	John, Mary	Gender + number	**John**
She	John, Mary	Gender + number	**Mary**
it	ball, park	Semantic compatibility	**Ball**
him	John, Mary, dog	Gender + semantic compatibility	**John**
they	John, Mary, dog	Plural + discourse coherence	**John + Mary + dog**

d) Final Coreference Chains

- **John → He → him → they**
- **Mary → She → they**
- **ball → it**
- **dog → they**

Rewritten Paragraph

John and Mary went to the park. John brought a ball. Mary wanted to play with the ball. The dog chased John excitedly. Finally, John, Mary, and the dog all went home.

e) Priority of Constraints

A suitable priority order is:

1. Gender and number agreement → 2. Semantic compatibility → 3. Coherence → 4. Recency

Agreement is strongest because an antecedent that does not match the pronoun's gender or number should normally be eliminated immediately. Semantic compatibility then removes unlikely interpretations. Coherence maintains a consistent entity chain, while recency is useful as a preference rather than an absolute rule.

If **recency is relaxed**, the system can still resolve references using gender, semantic compatibility, and discourse coherence. However, when several candidates satisfy the other constraints, the system may have more ambiguity because it no longer has recency to select the most likely antecedent.

2. Dialog Planning and Generation

Given Conversation

User: "I have an important exam tomorrow but I'm not able to concentrate."

Required Dialog Act: Advise + Encourage

a) Discourse Planning Steps

Before generating the response:

1. Identify the user's problem: difficulty concentrating.
2. Identify the important entity: **exam tomorrow**.
3. Select the dialog acts: **advice + encouragement**.
4. Establish a cause-effect relationship: difficulty concentrating → take a short break and refocus.
5. Include required keywords such as **focus, break, confident**.
6. Generate a polite and positive response.
7. Check that the response contains only **2–3 sentences**.

b) Three Possible Responses

Response 1:

Since your exam is tomorrow, take a short break to clear your mind and then return with better focus. You have prepared for this exam, so stay confident and concentrate on one topic at a time.

Response 2:

Your exam is important, so take a short break if you cannot concentrate and then focus on one small topic at a time. This will make studying easier, and you can feel confident about your preparation.

Response 3:

If you are struggling to concentrate for tomorrow's exam, take a short break and remove distractions before studying again. Keep your focus on one task at a time, and stay confident because steady preparation can help.

c) Best Response

Response 2 is the best choice.

Reasons:

- It clearly refers to the **exam** and **concentrate**.

- It uses **break, focus, and confident**.
- It provides practical advice.
- It also encourages the student.
- It has a clear **cause-effect relationship**.
- It is positive and logically consistent.
- It contains only **2 sentences**.

d) Effect of Violating Two Constraints

Violation 1 – Entity coherence:
If the response suddenly discusses unrelated topics instead of the exam and concentration problem, the conversation becomes confusing and less relevant.

Violation 2 – Positive tone:
A negative or discouraging response could increase the student's worry and fail to satisfy the **Encourage** dialog act.

3. Word Sense Disambiguation and Constraint-Based Translation

Given:

"The bank by the river flooded after the storm, but it was saved by quick action."

a) Word Sense Disambiguation

The correct meaning of **"bank"** is **riverbank**, not a financial institution.

Reason

The phrase **"by the river"** provides a strong contextual clue. Also, a riverbank can **flood**, whereas a financial institution is not normally described as flooding in this context.

Therefore:

bank → riverbank

This satisfies the WSD and semantic compatibility constraints.

b) Predicate Logic Representation

Let:

- RiverBank(b) = b is a riverbank.
- River(r) = r is a river.

- Storm(s) = s is a storm.
- Flood(b) = the riverbank flooded.
- Saved(b) = the riverbank was saved.
- QuickAction(a) = a is a quick action.
- After(s,b) = flooding occurred after the storm.
- SavedBy(b,a) = b was saved by action a.
- LocatedBy(b,r) = b is located by river r.

A simplified representation is:

RiverBank(b)

River(r)

Storm(s)

QuickAction(a)

LocatedBy(b,r)

Flood(b)

After(Flood(b), s)

SavedBy(b,a)

The contrast between the two clauses can be represented as:

Contrast(Flood(b), SavedBy(b,a))

This means the bank flooded, **but despite that, it was saved by quick action**.

c) Target Sentence – Simple English

The riverbank near the river flooded after the storm, but quick action saved it.

This preserves:

- The correct meaning of "bank"
- The river relationship
- The storm
- The flooding
- The saving action
- The contrast relationship

d) Simple RST-Style Discourse Tree

```
              Contrast
            /       \
           /         \
        Situation      Result
          |            |
    Riverbank flooded    Riverbank saved
    after the storm      by quick action
```

Or:

```
              Contrast
                 |

    ↓                    ↓
 Bank flooded after       Bank was saved
 the storm               by quick action
```

The word **“but”** signals the **Contrast** relation.

e) Advantages of Constraint-Based Approach

A constraint-based approach has several advantages over purely statistical machine translation:

Constraint-Based Approach	**Pure Statistical Approach**
Uses explicit linguistic constraints	Relies mainly on learned probabilities
Better WSD using context	May select an incorrect common meaning
Preserves important entities	May occasionally lose information
Maintains logical relationships	Logical relations may be weakened
Maintains discourse structure	Discourse relations may not always be preserved
Easier to explain decisions	Decisions can be harder to interpret

For this sentence, the constraint **“bank by the river”** strongly forces the meaning **riverbank**. The system also ensures that the **flooding and saving events refer to the same entity** and that the **contrast introduced by “but”** is preserved.

Overall Conclusion

Constraint-based NLP is useful when **accuracy, entity consistency, semantic correctness, and discourse preservation** are important. Statistical or neural approaches can provide strong general performance, but explicit constraints can prevent specific errors in linguistically important situations such as this ambiguous sentence.